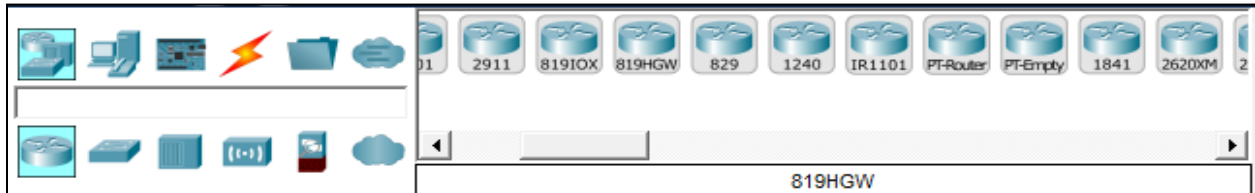


Practical No: 9

AIM: Using Packet Tracer ,Create a network with three routers with BGP and each router associated network will have minimum three PC.Show Connectivity

Step 1

From Network Devices → Routers, drag 3 routers (R1, R2, R3) to the workspace.



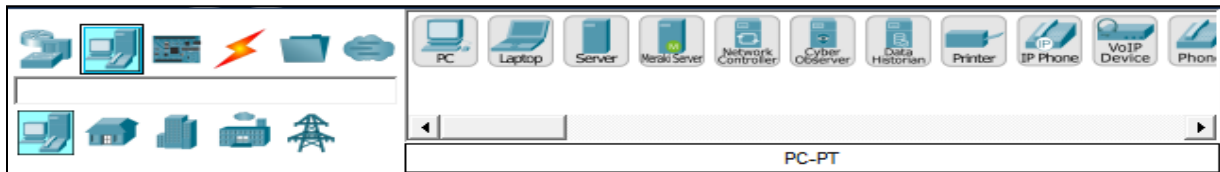
Step 2

Place **three switches**, one near each router.



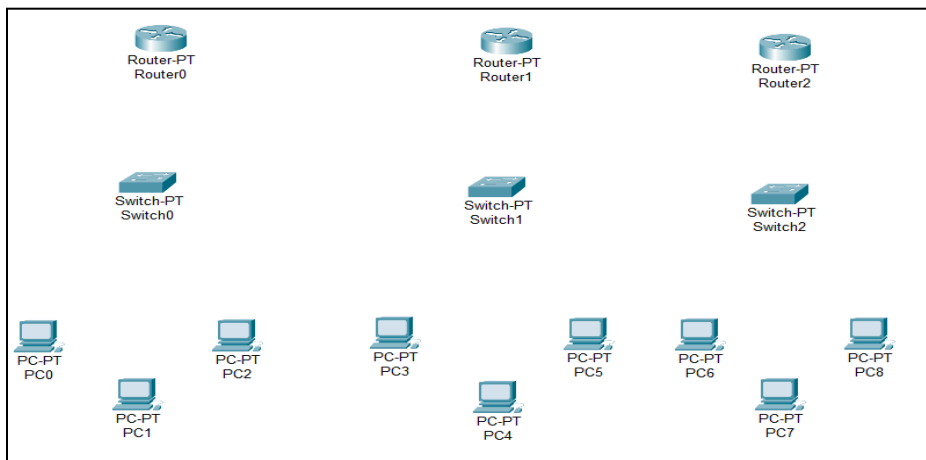
Step 3

From End Devices, drag **9 PCs** (3 PCs near each router).



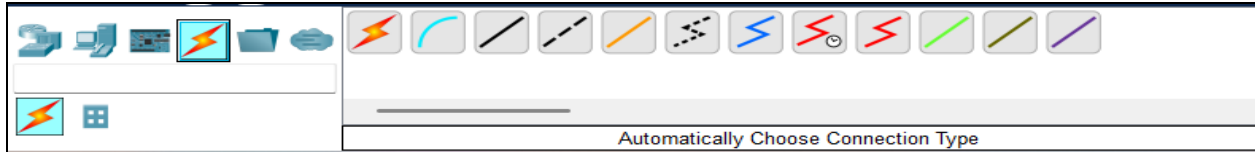
Step 4

Place routers at the top, switches below each router, and three PCs below each switch as shown.



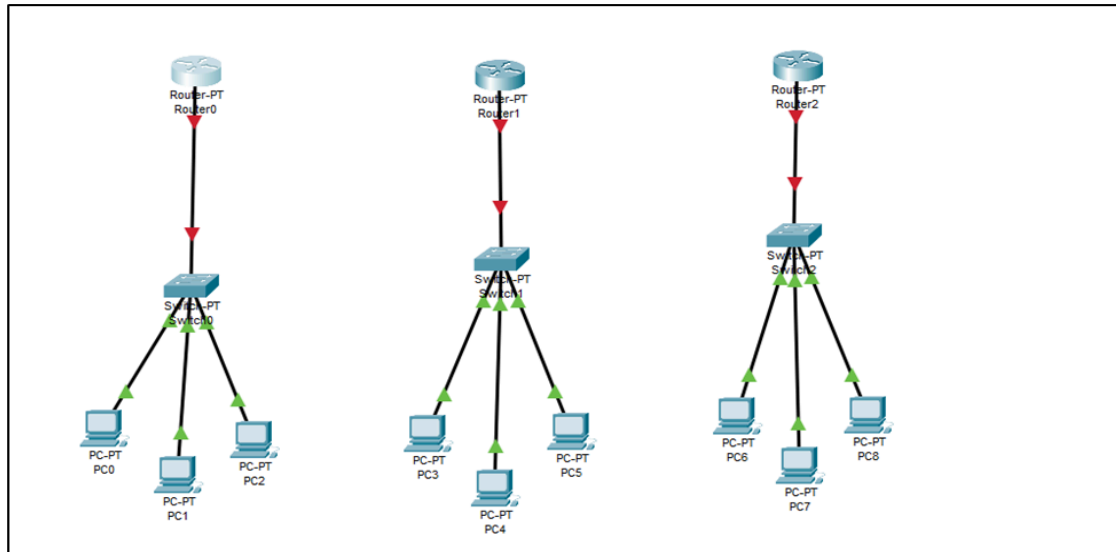
Step 5

From Connections, select **cable**.



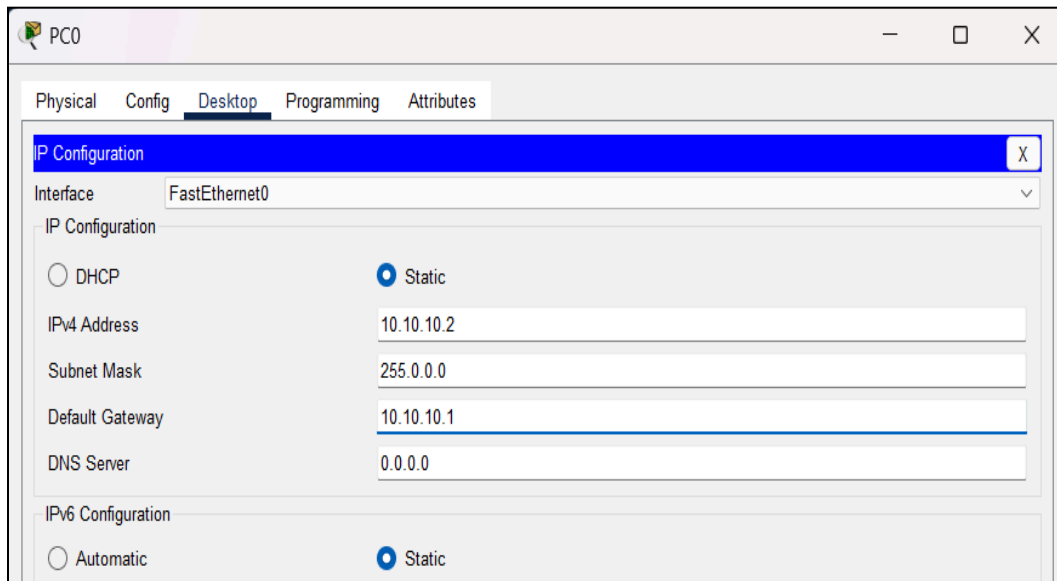
Step 6

Connect PCs to their respective router using FastEthernet ports.



Step 7

In the IP configuration window of PC0, set: IP Address, Default Gateway & Subnet Mask



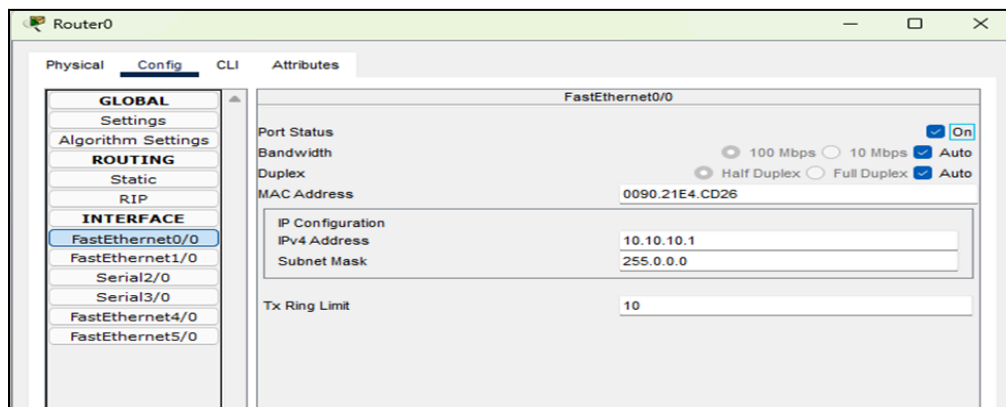
Step 8

Assign new IP address and default Gateway to all the PCs by following same procedure as PC0

	IP Configuration	Default Gateway
wPC0	10.10.10.2	10.10.10.1
PC1	10.10.10.3	10.10.10.1
PC2	10.10.10.4	10.10.10.1
PC3	20.20.20.2	20.20.20.1
PC4	20.20.20.3	20.20.20.1
PC5	20.20.20.4	20.20.20.1
PC6	30.30.30.2	30.30.30.1
PC7	30.30.30.3	30.30.30.1
PC8	30.30.30.4	30.30.30.1

Step 9

Click on **Router0** → **Config** → **Interfaces** and assign IP address and subnet mask.



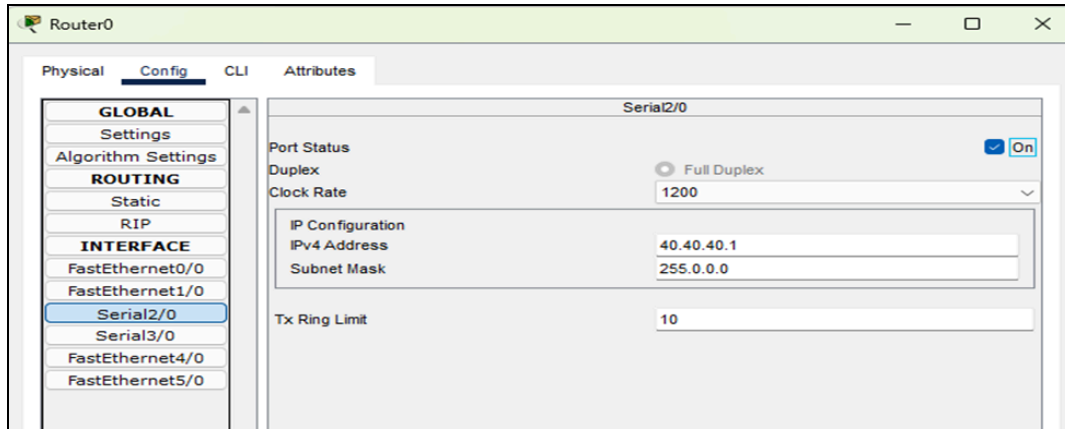
Step 10

Repeat the steps to all Routers and assign the IP address

	IP Configuration (FastEthernet 0/0)
Router0	10.10.10.1
Router1	20.20.20.1
Router2	30.30.30.1

Step 11

Configure the **Router0 Serial2/0** interface by turning the port **ON** and assigning the IP address and subnet mask



Step 12

Repeat the process for assigning new IP Configuration for the rest of the Routers. But in the middle Router you have to ON both **Serial2/0** and **Serial3/0** then assign a new IP address for both of them

	IP Configuration
Router0	Serial2/0 40.40.40.1
Router1	Serial2/0 40.40.40.2 Serial3/0 50.50.50.1
Router2	Serial2/0 50.50.50.2

Step 13

Open Router0→CLI and Configure BGP routing by adding the connected network and neighbor AS
Router0 – BGP Configuration (CLI Commands)

```
Router> enable
Router# configure terminal
Router(config)# router bgp 1
Router(config-router)# network 10.0.0.0
Router(config-router)# network 40.0.0.0
Router(config-router)# neighbor 40.40.40.2 remote-as 200
Router(config-router)# exit
```

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 40.40.40.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#network 10.0.0.0
Router(config-router)#network 40.0.0.0
Router(config-router)#neighbor 40.40.40.2 remote-as 2
Router(config-router)#exit
Router(config)#

```

Step 14
 Open Router1→CLI and Configure BGP routing by adding the connected network and neighbor AS
Router1 - BGP Configuration (CLI Commands)

```

Router> enable
Router# configure terminal
Router(config)# router bgp 2
Router(config-router)# network 20.0.0.0
Router(config-router)# network 40.0.0.0
Router(config-router)# network 50.0.0.0
Router(config-router)# neighbor 40.40.40.1 remote-as 1
Router(config-router)# exit

```

Step 15
 Open Router2→CLI and Configure BGP routing by adding the connected network and neighbor AS
Router1 - BGP Configuration (CLI Commands)

```

Router> enable
Router# configure terminal
Router(config)# router bgp 3
Router(config-router)# network 30.0.0.0
Router(config-router)# network 50.0.0.0
Router(config-router)# neighbor 50.50.50.1 remote-as 2
Router(config-router)# exit

```

Step 16
 Successful replies confirm that all PCs are connected through BGP-configured routers and the network is working properly.

Realtime									
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit Delete
Successful	PC0	PC2	ICMP		0.000	N	0	(e...	(delete)
Successful	PC0	PC3	ICMP		0.000	N	1	(e...	(delete)
Successful	PC1	PC5	ICMP		0.000	N	2	(e...	(delete)